

Computer Vision Project

Chess Piece Recognition System Using Few-Shot and Transfer Learning

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Goals



Integrated Detection System

Apply computer vision and deep learning to identify pieces across a full 64-square board.



Methodological Comparison

Evaluate Prototypical Networks (Few-Shot) vs. Fine-tuning (Transfer Learning) for piece classification.



Design Generalization

Extend recognition capabilities to non-standard, diverse chess piece designs.

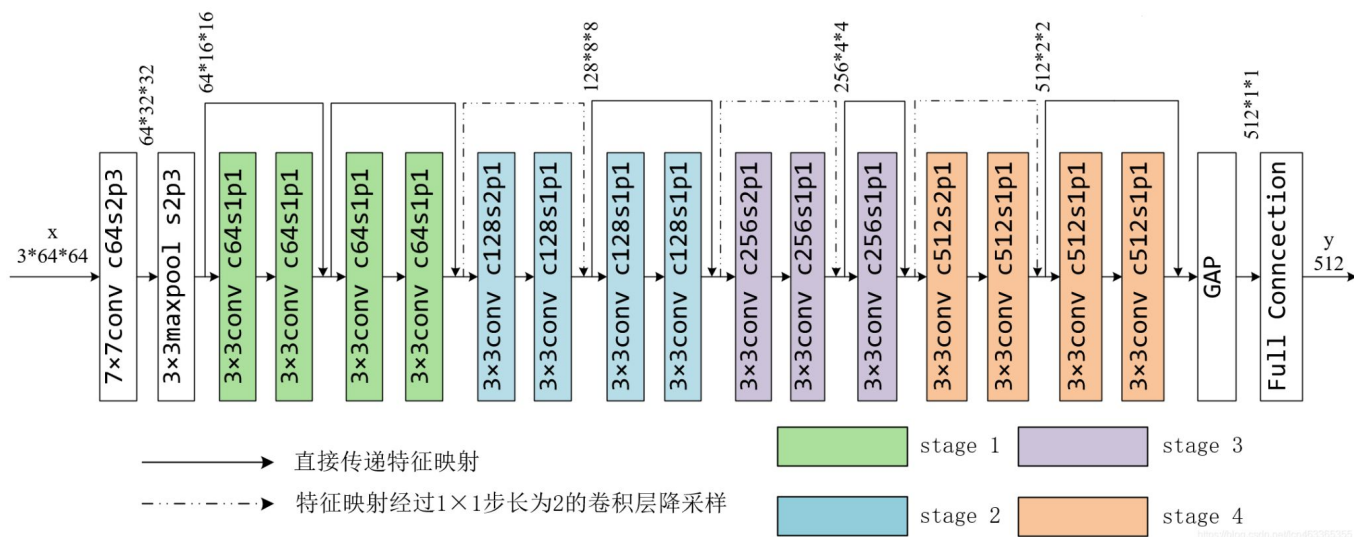


Outcome Analysis

Evaluation of both approaches, limitations, etc.

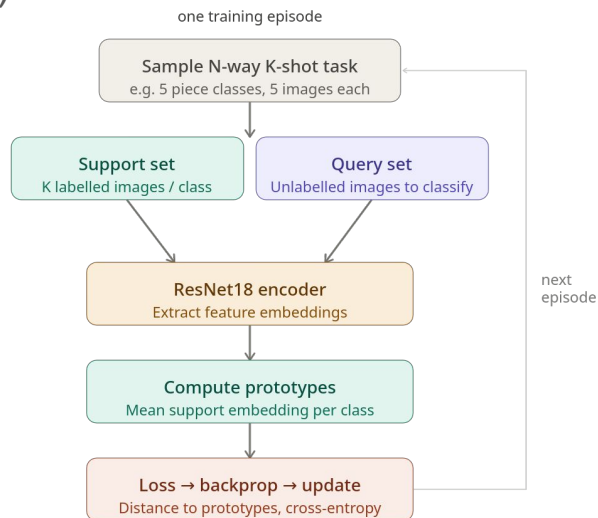
Methodology: Feature Extraction

- ResNet18 for feature extraction (512-dimensional embeddings).
- Trained on ~70k images for the 13 possible outputs (pieces + empty square).



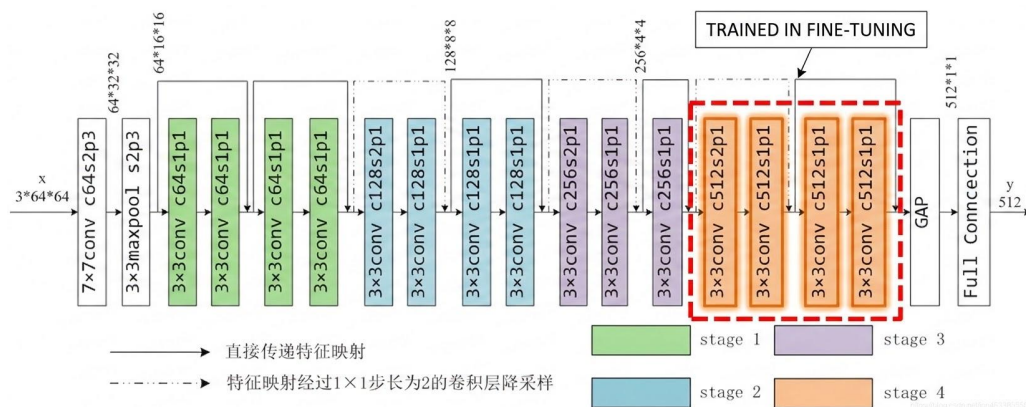
Methodology: Few-Shot Learning

- To allow for Few-Shot Learning, we used Episodic Training within a Prototypical Network architecture.
- The final NN is capable of classifying new pieces based on distance (Euclidean Distance) to the provided support set (Few-Shot).



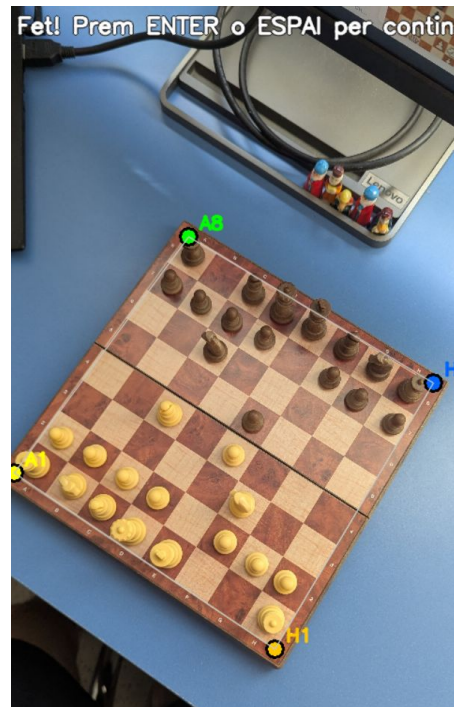
Methodology: Transfer Learning (Fine-Tuning)

- If the domain shifts significantly (e.g., different camera angles, lighting conditions, or piece shapes), we fine-tune the neural network by training only the fourth layer on the new data.
- This fine-tuning is lightweight, requiring only about 30 new images per class and 10 training epochs.



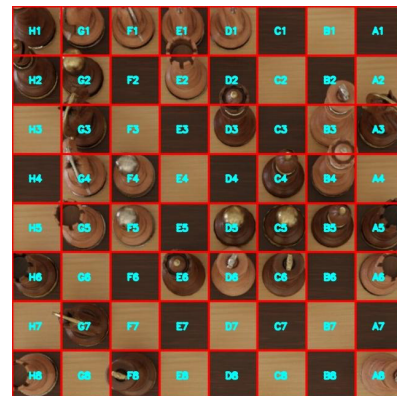
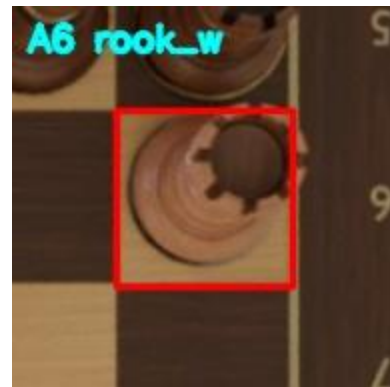
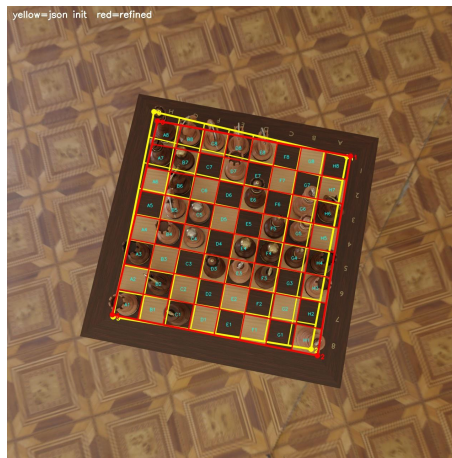
Methodology: Dataset

- **Training set:** ~70,000 synthetic piece images. Every photo includes a JSON file with the board corner coordinates and the piece type on each occupied square.
- **Custom evaluation set:** photographs taken with our own board, which has different chess pieces. Since no JSON is available, the user manually marks the 4 corners. To avoid labeling piece positions manually, we use the game's initial starting position — so the location of every piece is already known.
- **Generalization test:** we swapped the kings for non-standard figurines to evaluate whether the system can handle visually distinct, unconventional pieces.



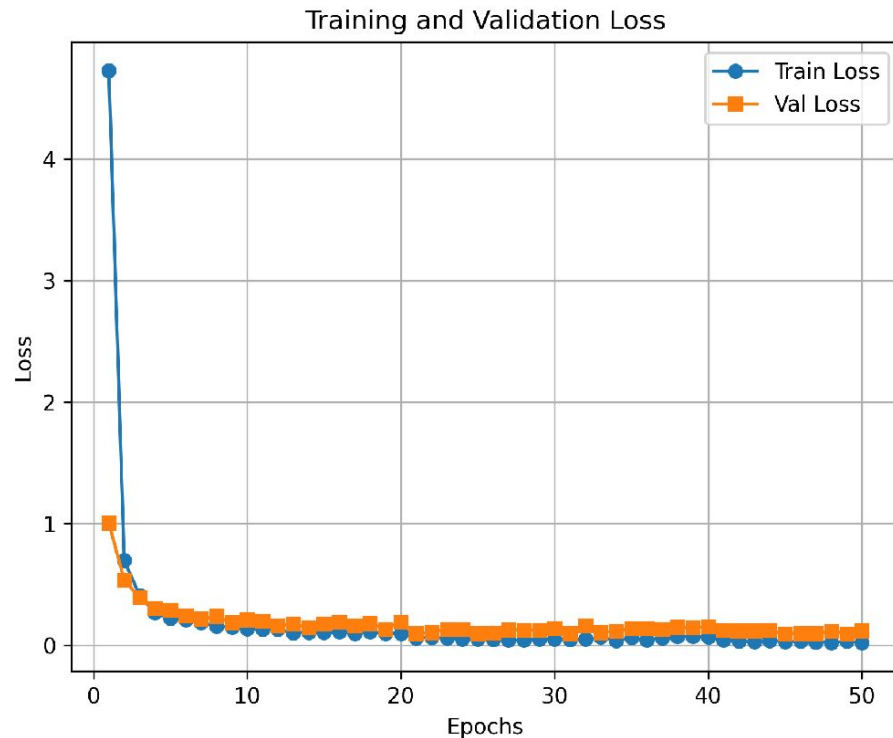
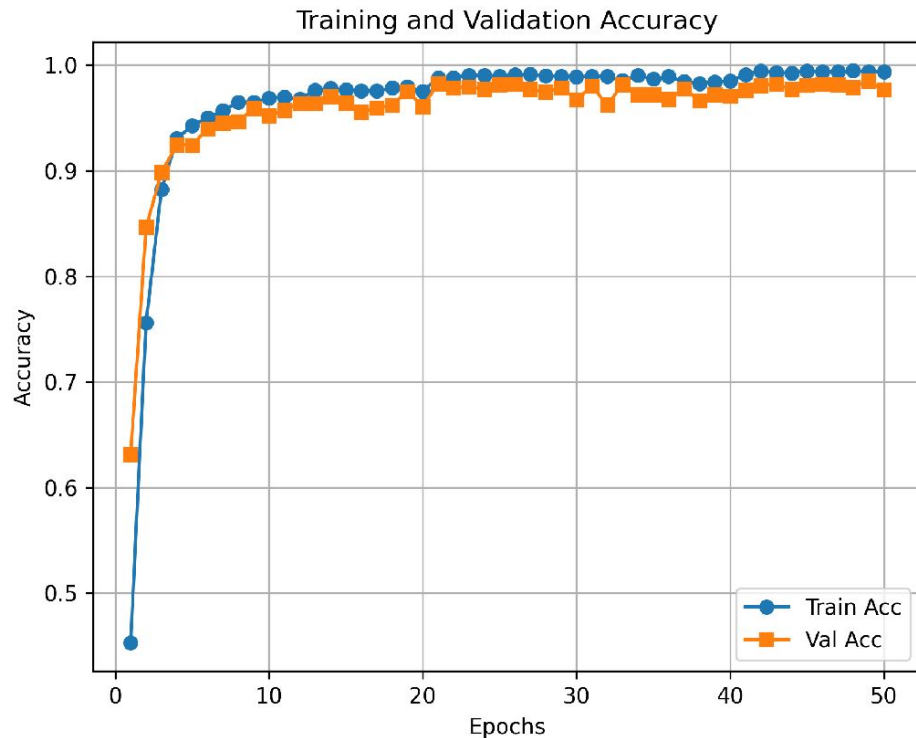
Methodology: Board Recognition

- User marks the 4 board corners to initialize detection; clicking order encodes board orientation
- Sobel edge detection refines each corner to subpixel accuracy
- Perspective correction via homography (warpPerspective) produces a square top-down view
- Board segmented into 64 individual squares; each piece cropped with a 1.4x expansion factor



Results: Training Curves

Accuracy and loss curves during feature extractor training on the synthetic dataset



Results: Synthetic Board

Predicted piece layout on a synthetic board alongside the digital FEN representation



Results: Accuracy Comparison

Classification accuracy: Few-Shot vs. Transfer Learning

Evaluated on 18 boards $\times$ 64 squares = 1,152 squares total

Few-Shot Learning

88.9%

128 errors / 1,152 squares

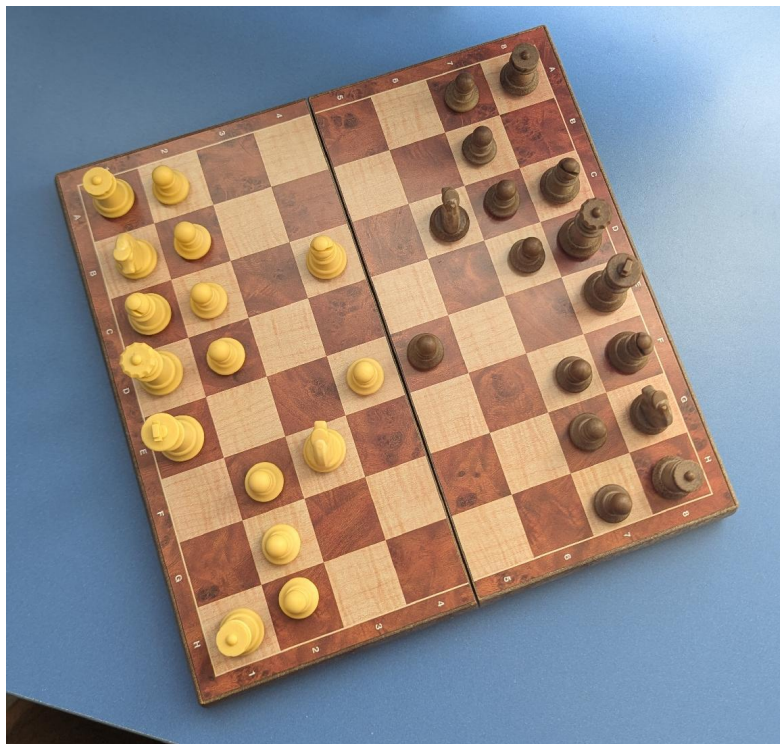
Transfer Learning

98.7%

15 errors / 1,152 squares

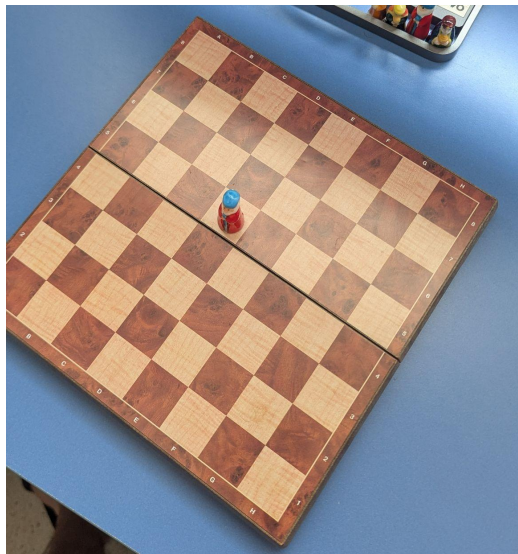
Results: Transfer Learning - Real Board

Recognition on a real chess board after fine-tuning (~30 images/class, 10 epochs)



Results: Custom Piece Generalization

System performance on non-standard king designs using the few-shot support set



Future Work & Conclusion

3D Modeling Integration

Incorporating depth sensors to mitigate occlusion issues in high-density board states.

Real-time AR Overlay

Developing mobile applications for live game analysis and move suggestion using the FEN output.

Distributed Learning

Cloud-based data collection to improve generalization across even more diverse custom chess sets.

Hyper-parameter Tuning

Optimizing the ResNet backbone specifically for low-contrast and grayscale photographic conditions.

Project Conclusion

The system demonstrates that transfer learning provides a robust foundation for chess piece recognition, while few-shot learning offers critical flexibility for rapidly adapting to new, non-standard designs.

Demo

